

# Telecom Customer Churn Prediction Using Machine Learning

## Project Overview

Customer churn is a critical problem in the telecom industry because losing customers directly impacts revenue and long-term growth. Companies need to identify customers who are likely to leave and take preventive actions to retain them. This project analyzes telecom customer data to understand the factors that contribute to churn and builds a machine learning model to predict whether a customer is likely to leave the company. Using data analysis and predictive modeling techniques, the project identifies important patterns in customer behavior and provides actionable business insights. The final outcome includes a predictive churn model and data visualizations that help business improve customer retention strategies.

## Objectives of the Project

- Understand customer demographics and service usage
- Perform data cleaning and preprocessing
- Conduct exploratory data analysis(EDA)
- Identify key factors influencing customer churn
- Build a machine learning model to predict churn
- Evaluate the model performance
- Create visual insights and dashboards for business decision making

## Tools & Technologies Used

- Python
- Pandas
- Numpy
- Matplotlib
- Seaborn
- Scikit-learn (Machine Learning)
- Jupyter Notebook
- Power BI (Dashboard & Visualization)

## Machine Learning Model

The project will implement a classification model using Logistic Regression to predict whether a customer will churn or stay with the company.

## Expected Outcome

The project will generate a predictive model capable of estimating customer churn probability and produce analytical insights that help telecom companies identify high-risk customers and improve retention strategies.

## Dataset Description

The dataset used in this project is the Telco Customer Churn dataset, which contains information about telecom customers and the services they use. The dataset includes customer demographic details, account information, service subscriptions, and billing data. The main objective is to analyze customer behavior and predict whether a customer is likely to churn.

## Dataset Information

- Total Records: 7043
- Total Features: 21

## Target Variable

The target variable in this dataset is **Churn**.

- Yes: Customer left the company
- No: Customer stayed with the company

## Key Features in the Dataset

- Customer Demographics (gender, SeniorCitizen, Partner, Dependents)
- Account Information (tenure, Contract, PaymentMethod)
- Services Used (InternetService, OnlineSecurity, StreamingTV, etc.)
- Billing Information (MonthlyCharges, TotalCharges) These features help identify patterns and factors that influence customer churn.

## Importing Required Libraries

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

# Loading the Dataset

```
In [4]: import pandas as pd
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
In [5]: df.head()
```

Out[5]:

|  | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | Multipl |
|--|------------|--------|---------------|---------|------------|--------|--------------|---------|
|--|------------|--------|---------------|---------|------------|--------|--------------|---------|

|   |            |        |   |     |    |    |     |    |
|---|------------|--------|---|-----|----|----|-----|----|
| 0 | 7590-VHVEG | Female | 0 | Yes | No | 1  | No  | No |
| 1 | 5575-GNVDE | Male   | 0 | No  | No | 34 | Yes |    |
| 2 | 3668-QPYBK | Male   | 0 | No  | No | 2  | Yes |    |
| 3 | 7795-CFOCW | Male   | 0 | No  | No | 45 | No  | No |
| 4 | 9237-HQITU | Female | 0 | No  | No | 2  | Yes |    |

5 rows × 21 columns



## Data Inspection

```
In [6]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   customerID            7043 non-null   object
 1   gender                7043 non-null   object
 2   SeniorCitizen         7043 non-null   int64
 3   Partner               7043 non-null   object
 4   Dependents            7043 non-null   object
 5   tenure                7043 non-null   int64
 6   PhoneService          7043 non-null   object
 7   MultipleLines         7043 non-null   object
 8   InternetService       7043 non-null   object
 9   OnlineSecurity        7043 non-null   object
10   OnlineBackup          7043 non-null   object
11   DeviceProtection      7043 non-null   object
12   TechSupport           7043 non-null   object
13   StreamingTV           7043 non-null   object
14   StreamingMovies       7043 non-null   object
15   Contract              7043 non-null   object
16   PaperlessBilling      7043 non-null   object
17   PaymentMethod         7043 non-null   object
18   MonthlyCharges        7043 non-null   float64
19   TotalCharges          7043 non-null   object
20   Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

```
In [7]: df.isnull().sum()
```

```

Out[7]: customerID            0
        gender              0
        SeniorCitizen       0
        Partner             0
        Dependents          0
        tenure              0
        PhoneService        0
        MultipleLines       0
        InternetService     0
        OnlineSecurity      0
        OnlineBackup        0
        DeviceProtection    0
        TechSupport         0
        StreamingTV        0
        StreamingMovies     0
        Contract            0
        PaperlessBilling    0
        PaymentMethod       0
        MonthlyCharges      0
        TotalCharges        0
        Churn               0
dtype: int64

```

```
In [8]: df[df["TotalCharges"] == ""]
```

Out[8]: **customerID gender SeniorCitizen Partner Dependents tenure PhoneService Multiple**

0 rows × 21 columns



```
In [9]: df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
In [10]: df.dtypes
```

```
Out[10]: customerID      object
gender      object
SeniorCitizen  int64
Partner      object
Dependents    object
tenure      int64
PhoneService  object
MultipleLines object
InternetService object
OnlineSecurity object
OnlineBackup  object
DeviceProtection object
TechSupport   object
StreamingTV   object
StreamingMovies object
Contract      object
PaperlessBilling object
PaymentMethod object
MonthlyCharges float64
TotalCharges  float64
Churn         object
dtype: object
```

```
In [11]: df.isnull().sum()
```

```
Out[11]: customerID      0
gender      0
SeniorCitizen  0
Partner      0
Dependents    0
tenure      0
PhoneService  0
MultipleLines  0
InternetService  0
OnlineSecurity  0
OnlineBackup  0
DeviceProtection  0
TechSupport    0
StreamingTV    0
StreamingMovies  0
Contract      0
PaperlessBilling  0
PaymentMethod  0
MonthlyCharges  0
TotalCharges  11
Churn         0
dtype: int64
```

```
In [12]: df = df.dropna()
```

```
In [13]: df.shape
```

```
Out[13]: (7032, 21)
```

## Exploratory Data Analysis(EDA)

### Churn Distribution

```
In [14]: df["Churn"].value_counts()
```

```
Out[14]: Churn
No      5163
Yes     1869
Name: count, dtype: int64
```

```
In [15]: df["Churn"].value_counts(normalize=True)*100
```

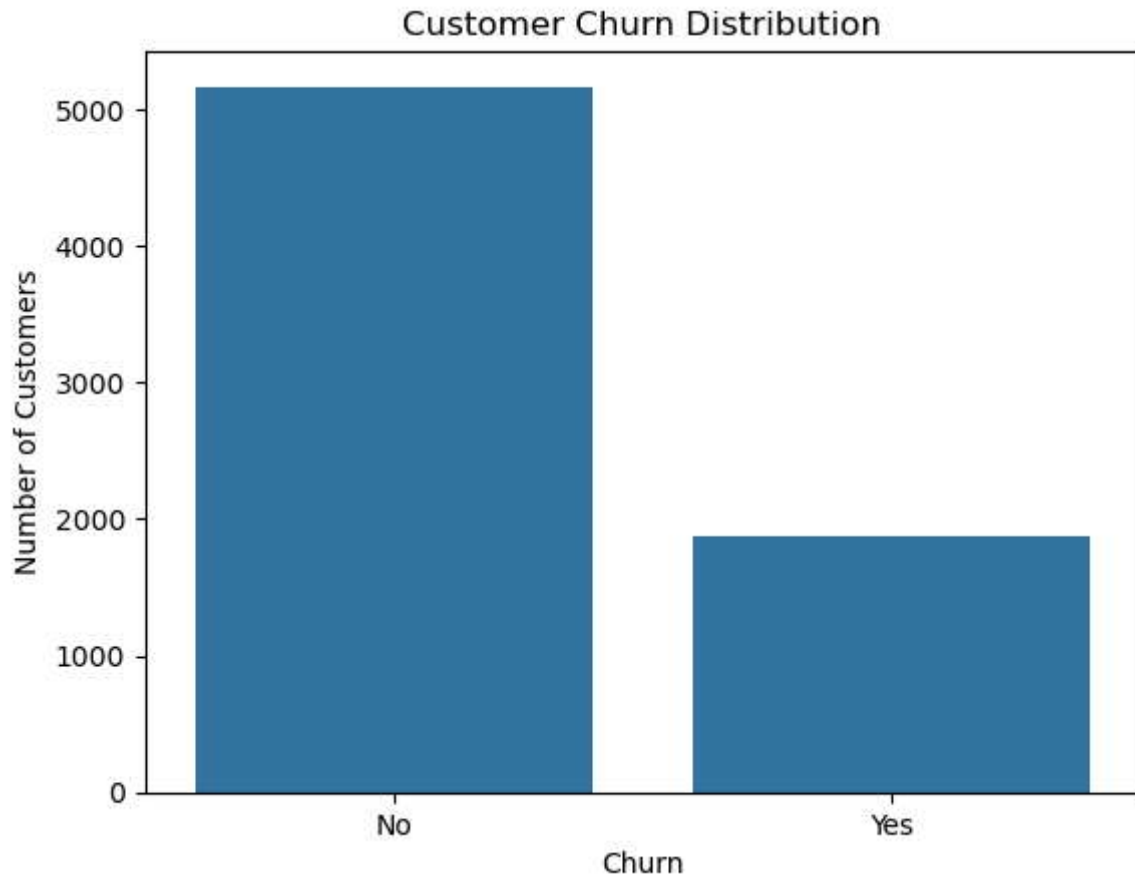
```
Out[15]: Churn
No      73.421502
Yes     26.578498
Name: proportion, dtype: float64
```

### Insight

Approximately 26.6% of customers have churned while 73.4% remain with the company. This indicates that customer churn is a significant issue for the business, and understanding the

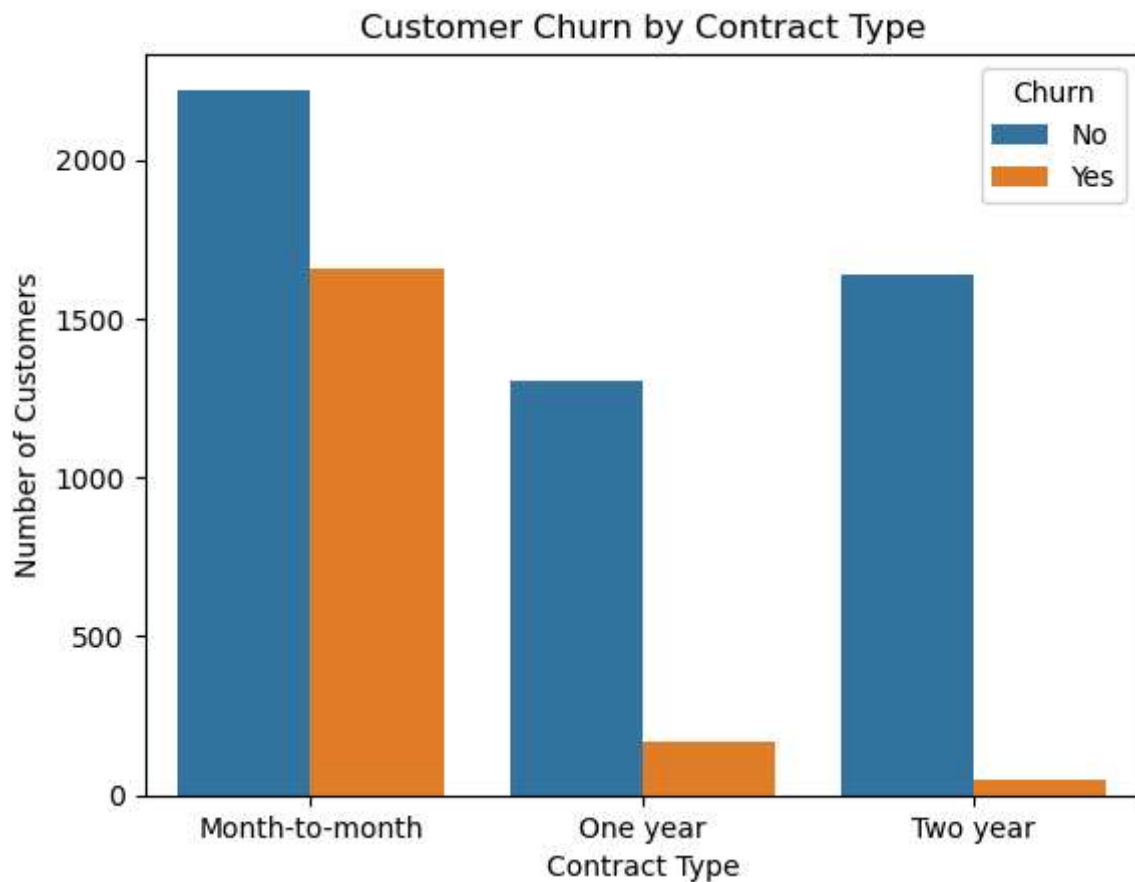
factors influencing churn can help the company take preventive actions.

```
In [16]: import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x="Churn", data=df)
plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")
plt.show()
```



## Churn by Contract Type

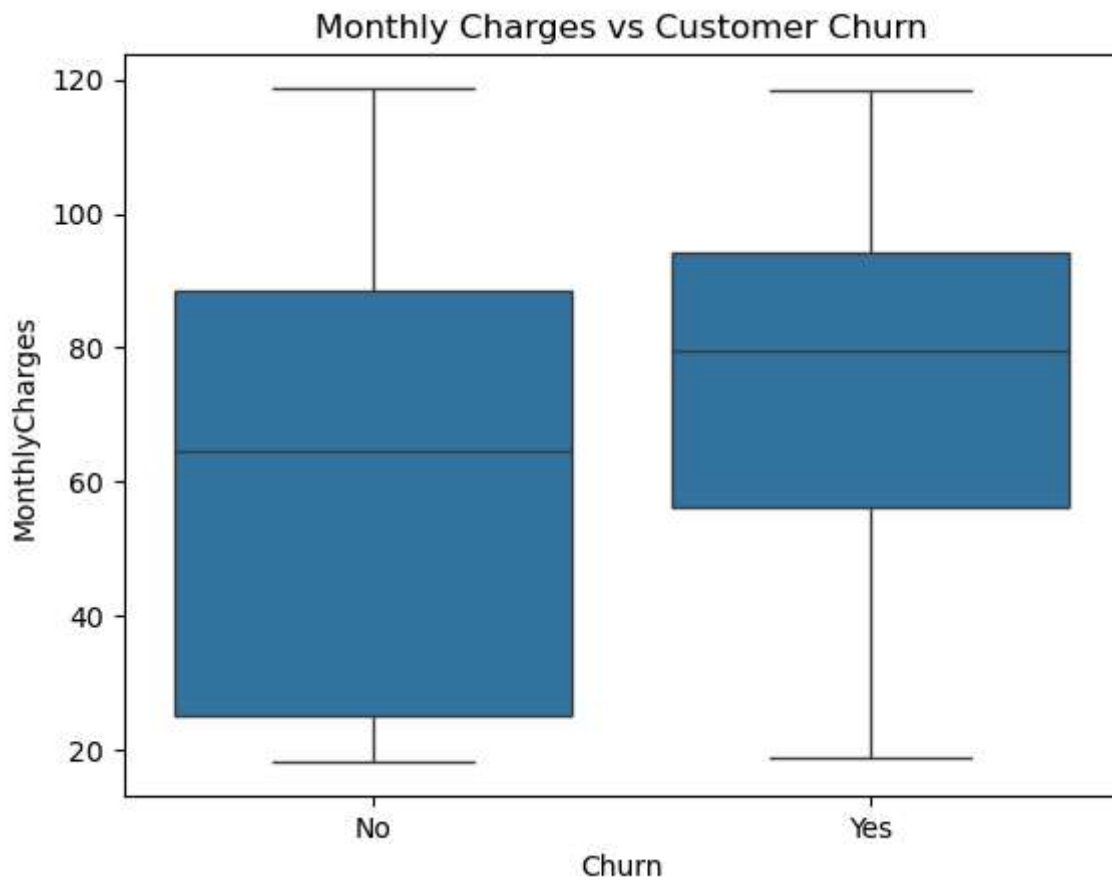
```
In [17]: import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x="Contract", hue="Churn", data=df)
plt.title("Customer Churn by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
plt.show()
```



## Insight

Customers with month-to-month contracts show the highest churn rate compared to customers with one-year or two-year contracts. This suggests that customers with short-term contracts are more likely to leave the service because they have more flexibility to switch providers. Encouraging customers to move toward longer-term contracts may help reduce churn.

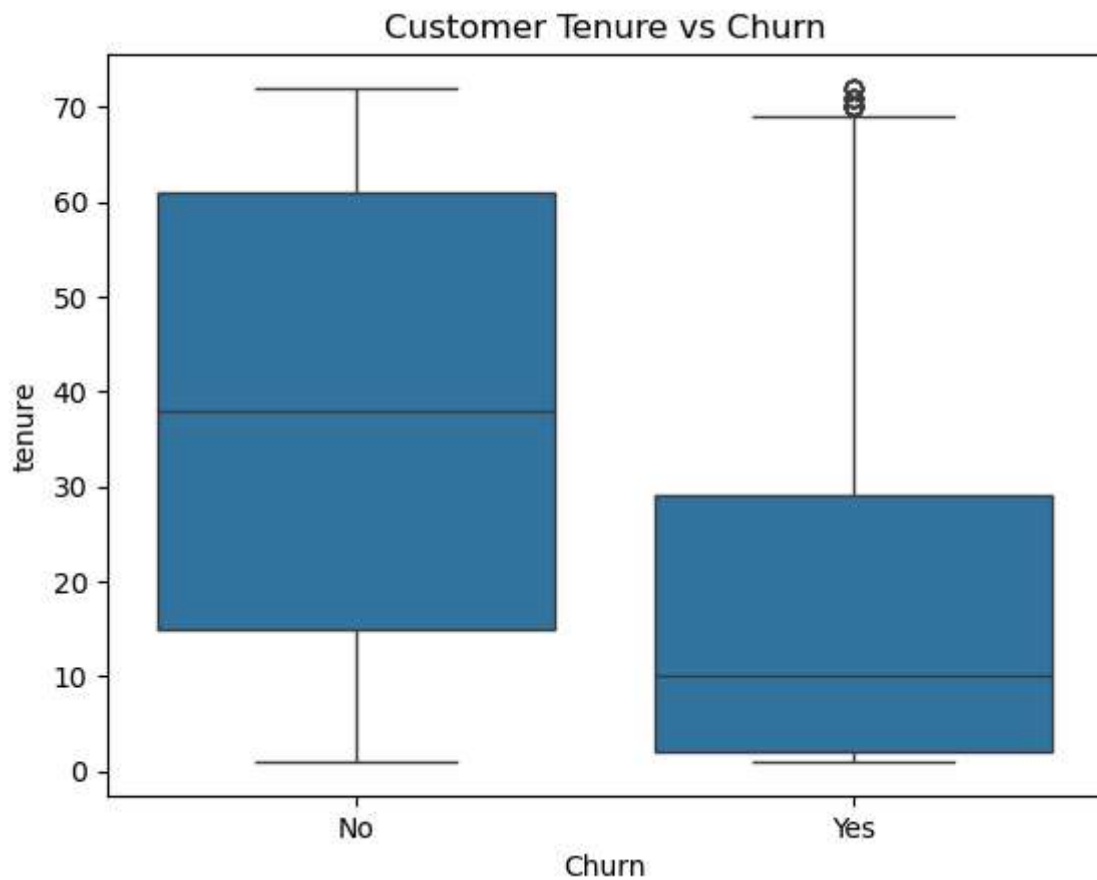
```
In [18]: sns.boxplot(x="Churn",y="MonthlyCharges",data=df)
plt.title("Monthly Charges vs Customer Churn")
plt.show()
```



## Insight

The boxplot shows that customers who churn generally have slightly higher monthly charges compared to customers who stay. Although there is overlap between the two groups, the median monthly charge for churned customers appears higher. This suggests that higher service costs may contribute to customer churn.

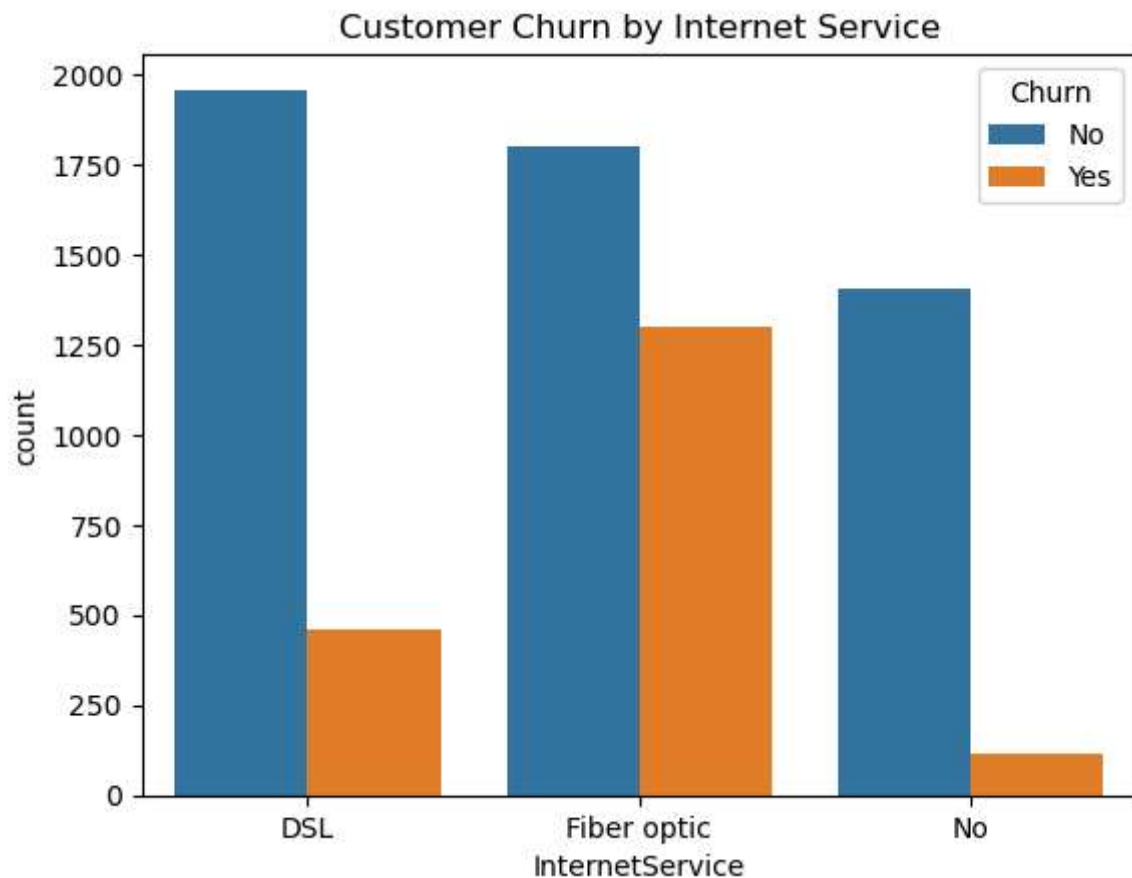
```
In [19]: sns.boxplot(x="Churn",y="tenure",data=df)
plt.title("Customer Tenure vs Churn")
plt.show()
```



## Insight

The boxplot shows that customers who churn typically have lower tenure compared to customers who stay. Most churned customers leave within the early months of service, while customers with longer tenure tend to remain with the company. This suggests that improving the experience of new customers could help reduce churn.

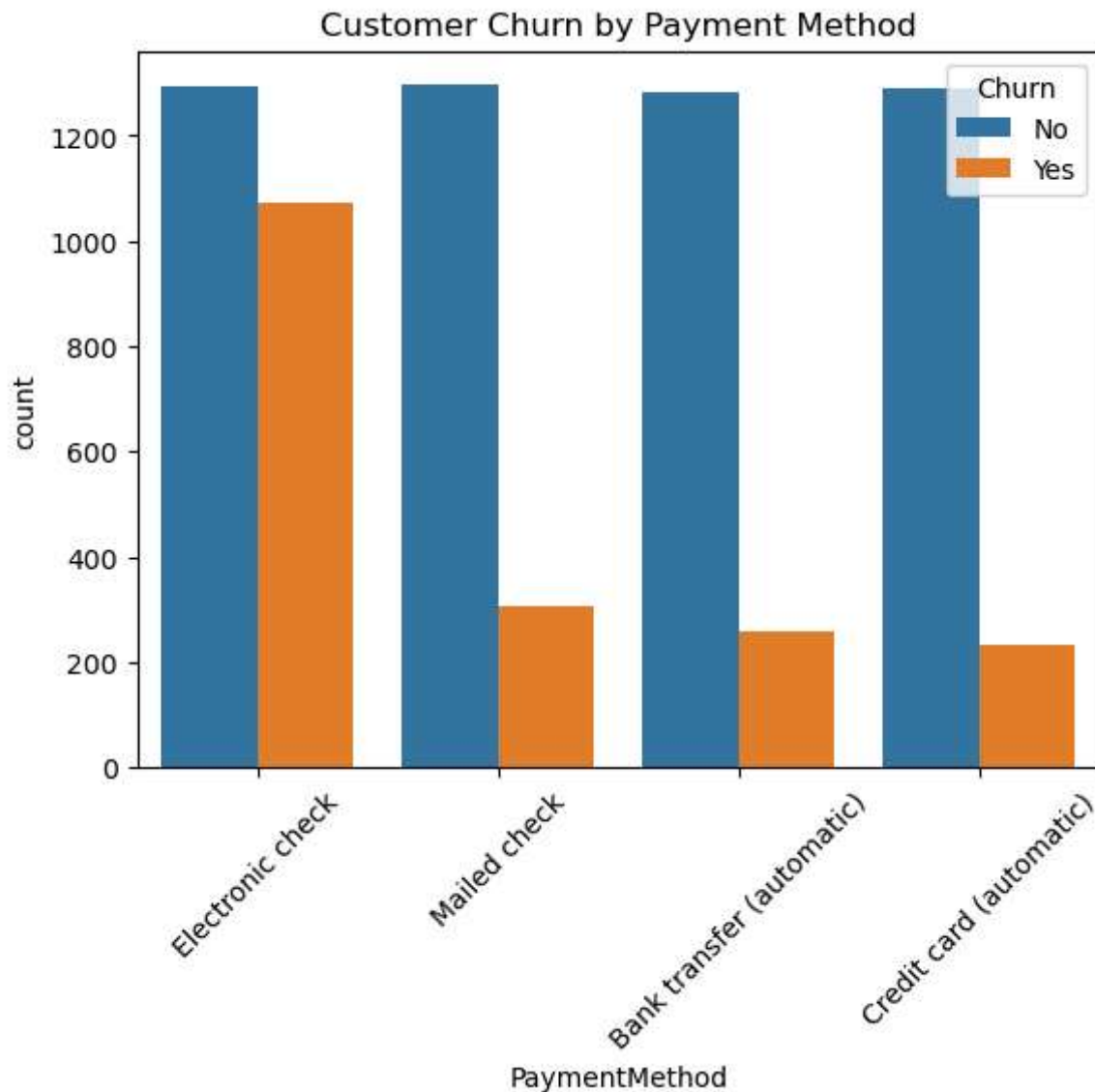
```
In [20]: sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Customer Churn by Internet Service")
plt.show()
```



## Insight

Customers using Fiber Optic internet service show the highest churn compared to DSL and customers without internet service. This may indicate that Fiber Optic users experience higher service costs or service quality issues, leading to higher churn rates. Improving service quality and pricing strategies for Fiber Optic customers may help reduce churn.

```
In [21]: sns.countplot(x="PaymentMethod",hue="Churn",data=df)
plt.title("Customer Churn by Payment Method")
plt.xticks(rotation=45)
plt.show()
```



## Insight

Customers using electronic check as their payment method show the highest churn compared to other payment methods. Customers using automatic payment methods such as credit card or bank transfer tend to churn less. Encouraging customers to switch to automatic payment methods may help reduce churn.

## Feature Engineering

This step prepares the dataset for machine learning by converting categorical variables into numerical form and selecting relevant features for the model.

```
In [22]: df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})
```

```
In [23]: df["Churn"].head()
```

```
Out[23]: 0    0
          1    0
          2    1
          3    0
          4    1
          Name: Churn, dtype: int64
```

```
In [24]: df=df.drop("customerID",axis=1)
```

```
In [25]: df_encoded = pd.get_dummies(df,drop_first=True)
```

```
In [26]: df_encoded.head()
```

```
Out[26]:
```

|   | SeniorCitizen | tenure | MonthlyCharges | TotalCharges | Churn | gender_Male | Partner_Yes |
|---|---------------|--------|----------------|--------------|-------|-------------|-------------|
| 0 | 0             | 1      | 29.85          | 29.85        | 0     | False       | True        |
| 1 | 0             | 34     | 56.95          | 1889.50      | 0     | True        | False       |
| 2 | 0             | 2      | 53.85          | 108.15       | 1     | True        | False       |
| 3 | 0             | 45     | 42.30          | 1840.75      | 0     | True        | False       |
| 4 | 0             | 2      | 70.70          | 151.65       | 1     | False       | False       |

5 rows × 31 columns



```
In [27]: df_encoded.shape
```

```
Out[27]: (7032, 31)
```

## Feature Engineering Summary

Categorical variables were converted into numerical variables using one-hot encoding. This process created additional columns representing different categories of the original variables. The dataset now contains 7032 rows and 31 columns, making it suitable for machine learning algorithms.

```
In [28]: x = df_encoded.drop("Churn",axis=1)
          y = df_encoded["Churn"]
```

```
In [29]: from sklearn.model_selection import train_test_split
          x_train,x_test,y_train,y_test = train_test_split(
              x,y,test_size=0.2,random_state=42
          )
```

```
In [30]: x_train.shape
```

```
Out[30]: (5625, 30)
```

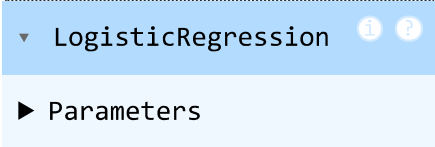
```
In [31]: x_test.shape
```

```
Out[31]: (1407, 30)
```

## Machine Learning Model

### Logistic Regression Model for Customer Churn Prediction

```
In [32]: from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=5000)
model.fit(x_train,y_train)
```

```
Out[32]: 
  ▾ LogisticRegression ⓘ ?
    ▶ Parameters
```

```
In [33]: y_pred = model.predict(x_test)
```

```
In [34]: from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test,y_pred)
print("Model Accuracy:",accuracy)
```

Model Accuracy: 0.7874911158493249

### Model Accuracy

The Logistic Regression model achieved an accuracy of approximately **78.7%** on the test dataset. This means the model correctly predicts whether a customer will churn or not in about 79 out of 100 cases.

```
In [35]: from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test,y_pred)
print(cm)
```

```
[[915 118]
 [181 193]]
```

### Confusion Matrix Interpretation

The confusion matrix shows that the model correctly predicted 915 non-churn customers and 193 churn customers. However, the model incorrectly predicted 118 customers as churn when they did not churn, and failed to identify 181 customers who actually churned. Overall, the model demonstrates reasonable performance with an accuracy of approximately 78.7%

```
In [36]: from sklearn.metrics import classification_report
print(classification_report(y_test,y_pred))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.83      | 0.89   | 0.86     | 1033    |
| 1            | 0.62      | 0.52   | 0.56     | 374     |
| accuracy     |           |        | 0.79     | 1407    |
| macro avg    | 0.73      | 0.70   | 0.71     | 1407    |
| weighted avg | 0.78      | 0.79   | 0.78     | 1407    |

## Model Evaluation Metrics

The classification report provides precision, recall, and F1-score for the churn prediction model. Precision indicates how many predicted churn customers actually churned, while recall measures how many actual churn customers were correctly identified by the model. The F1-score balances both precision and recall and provides an overall measure of the model's performance. These metrics help evaluate the effectiveness of the churn prediction model beyond simple accuracy.

## Classification Report Interpretation

The Logistic Regression model achieved an overall accuracy of approximately 79%. For non-churn customers (class0), the model shows strong performance with a precision of 0.83 and recall of 0.89. For churn customers (class1), the model achieved a precision of 0.62 and recall 0.52, indicating moderate ability to identify customers likely to churn. The macro and weighted averages provide an overall summary of model performance across both classes.

```
In [37]: df.to_csv("telco_churn_cleaned.csv", index=False)
```

```
In [38]: import os
os.listdir()
```

```

Out[38]: ['.anaconda',
          '.conda',
          '.continuum',
          '.idlerc',
          '.ipynb_checkpoints',
          '.ipython',
          '.jupyter',
          '.matplotlib',
          '.vscode',
          'anaconda3',
          'anaconda_projects',
          'AppData',
          'Application Data',
          'Contacts',
          'Cookies',
          'demos.txt',
          'Desktop',
          'Documents',
          'Downloads',
          'E-commerce_Sales_Analysis.ipynb',
          'Favorites',
          'Links',
          'Local Settings',
          'Microsoft',
          'Music',
          'My Documents',
          'NetHood',
          'NTUSER.DAT',
          'ntuser.dat.LOG1',
          'ntuser.dat.LOG2',
          'NTUSER.DAT{833617ac-a8ec-11f0-9674-ad90336954e5}.TM.blf',
          'NTUSER.DAT{833617ac-a8ec-11f0-9674-ad90336954e5}.TMContainer000000000000000000
1.regtrans-ms',
          'NTUSER.DAT{833617ac-a8ec-11f0-9674-ad90336954e5}.TMContainer000000000000000000
2.regtrans-ms',
          'ntuser.ini',
          'OneDrive',
          'online_retail_II.csv',
          'practice.txt',
          'PrintHood',
          'Recent',
          'Saved Games',
          'Searches',
          'SendTo',
          'Start Menu',
          'telco_churn_cleaned.csv',
          'Telco_Customer_Churn_Analysis.ipynb',
          'Templates',
          'Videos',
          'WA_Fn-UseC_-Telco-Customer-Churn.csv']

```

```

In [39]: import os
          os.getcwd()

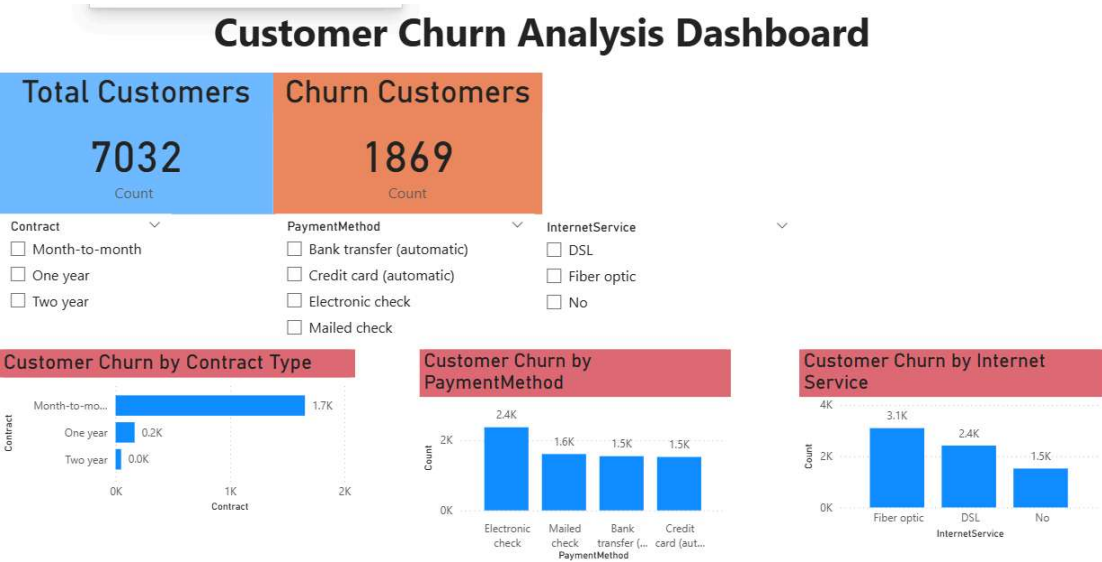
```

```

Out[39]: 'C:\\Users\\anura'

```

# Power BI Dashboard



1. Customers with month-to-month contracts have the highest churn rate.
2. Customers using Fiber Optic internet service churn more compared to DSL and customers with no internet service.
3. Customers who use Electronic Check as their payment method show the highest churn.
4. Customers with longer tenure are less likely to churn.

## Conclusion

This project analyzed customer churn behavior in a telecom company using data analytics and machine learning techniques. The dataset was cleaned and explored using Python libraries such as Pandas, Numpy, and Matplotlib. Exploratory Data Analysis was performed to identify patterns and relationships between customer characteristics and churn behavior. A Logistic Regression machine learning model was built to predict customer churn. The model achieved an accuracy of approximately 78%, which indicates a reasonably good predictive performance for identifying customers likely to leave the service. A Power BI dashboard was created to visually present the key findings of the analysis. The dashboard highlights important factors influencing churn such as contract type, internet service, and payment method. The insights from this analysis can help telecom companies design better retention strategies and reduce customer churn.

In [ ]: